

# ZAFEER AHMED QADEER

Mechanical Engineer | M.S. NYU Tandon '26 | Product Design · Manufacturing & Process · HVAC/MEP | SolidWorks CSWA · GD&T · Revit MEP  
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## PROFESSIONAL SUMMARY

Mechanical engineer with an M.S. from NYU Tandon (GPA 3.73, May 2026), working across mechanical product design, manufacturing and process engineering, and building HVAC/MEP systems. Co-founded Ingentas Robotics and delivered 8 client products to paying customers, \$7,000+ in engineering revenue, every one on time. At a tooling manufacturer I traced recurring mold defects back to gaps in dimensional inspection, revised those checkpoints, and cut rework 10-15%. Comfortable from concept and CAD through tolerance analysis, simulation, process selection, fabrication and acceptance testing. First-author publication on LiDAR and AI in drones, IOSR-JMCE, April 2024. Authorized to work in the U.S.

## TECHNICAL SKILLS

**CAD & Mechanical Design:** SolidWorks (CSWA), AutoCAD, Fusion 360, CATIA, Solid & Surface Modeling, Large Assembly Modeling, GD&T (ASME Y14.5), Tolerance Stack-up Analysis, DFM, DFA, DFMEA, Sheet Metal Design, Engineering Drawings, BOM Management

**Manufacturing & Fabrication:** Injection Molding, Blow Molding, Die Casting, Rotational Molding, CNC Machining, Additive Manufacturing, FDM, SLA, Rapid Prototyping, Hand Prototyping, Tooling Design, Fixture Design, Material Selection, Vendor Sourcing, Low-Volume Manufacturing

**Process & Quality:** Process Optimization, Root Cause Analysis, Continuous Improvement, Dimensional Inspection, Standard Work Instructions, Statistical Process Control, FMEA, Lean Manufacturing, 5S, Poka-Yoke, ISO 9001, Design Validation, Takt Time, QA Documentation

**Simulation & Analysis:** ANSYS (Structural and Thermal FEA), SolidWorks Flow Simulation (certified), Computational Fluid Dynamics, MATLAB, Thermal Analysis, Heat Transfer, Fluid Mechanics, Kinematic Analysis, Statics

**Electromechanical & Embedded:** Sensor Integration (LiDAR, ultrasonic, camera, FPV), Motor, ESC and Controller Packaging, Wiring and Cable Routing, Arduino (Uno, MEGA), Firmware Configuration, C, C++

**HVAC & MEP Systems:** HVAC System Design, Cooling & Heating Load Calculations, Duct & Pipe Sizing, Psychrometrics, VRF/VRV Systems, MEP Coordination, BIM Coordination, Revit MEP, AutoCAD MEP, Navisworks, Construction Documents, Submittal Review, Commissioning, As-Built Verification

**Codes & Standards:** ASHRAE 62.1, ASHRAE 90.1, IMC, NFPA 90A/90B, NYCECC, Local Law 97

## EDUCATION

### New York University - Tandon School of Engineering

May 2026

M.S. Mechanical Engineering | GPA 3.73/4.0 | New York, NY

Coursework: New Product Development, Additive Manufacturing of Metallic Materials, Security in Additive Manufacturing, CFD & Heat Transfer, Thermal Engineering Fundamentals, Energy Project Financing, Linear Control Theory, Solid Mechanics

### Muffakham Jah College of Engineering & Technology (Osmania University)

Jun 2024

B.E. Mechanical Engineering | Hyderabad, India

Coursework: Design for Manufacturing, Mechanics of Materials, Thermal Engineering, Fluid Mechanics

## PROFESSIONAL EXPERIENCE

### NYU Hyperloop Team, NYU Tandon

Jan 2026 – May 2026

Mechanical Design Engineer | New York, NY

- Held mechanical routing and installation clearances against the electrical and structural subsystems on a 20+ member team, resolving the interferences in design reviews before anything reached fabrication.
- Applied GD&T (ASME Y14.5) and DFM across 30+ custom pod components, releasing drawings cleared for fabrication.
- Maintained the engineering BOM, assembly drawings and design rationale the team worked from across sprint reviews.

### Sun Max Solar Systems

Oct 2023 – Jan 2024

Solar Energy Engineering Intern | Hyderabad, India

- Ran site surveys and commissioning walkthroughs across residential and commercial PV installations, checking as-built conditions against design drawings and writing the handover documentation clients signed off.
- Performed QA on lithium-ion and lead-acid battery systems and coordinated installations against local regulatory requirements.

### Advance Tooling Systems

Jun 2023 – Jul 2023

Manufacturing Engineering Intern, Tooling & Manufacturing | Hyderabad, India

- Traced recurring mold fitment defects back to gaps in dimensional inspection and revised those checkpoints, cutting rework 10-15%, saving 1.5-2 hours per production cycle, and improving quality on customer consignments by 40%.
- Selected polymers and alloys for mold fabrication across 50+ tooling units spanning injection molding, blow molding, die casting and rotational molding, and evaluated draft, wall thickness and tooling feasibility on each.
- Kept the QA records for every unit inspected against supplied blueprints.

### Mahindra & Mahindra Limited (Global Automotive OEM)

May 2023 – Jun 2023

Manufacturing Engineering Intern, Automotive & Farm Equipment | Zaheerabad, India

- Mapped a multi-stage tractor assembly line, documenting station-level quality gates, takt time and material flow for the plant engineering team.
- Ran in-process quality inspections on the line, checking component fitment, torque specifications and dimensional tolerances against engineering drawings.

### **Ingentas Robotics (Engineering Startup)**

Jan 2022 – Sep 2022

Mechanical Design & Manufacturing Engineer | Co-Founder | Hyderabad, India

- Co-founded the firm and took 8 products to paying customers: a UAV quadcopter, a Bluetooth-controlled RC vehicle, an ultrasonic-sensing vehicle and a line-following autonomous vehicle. \$7,000+ in engineering revenue over 9 months, every one delivered on time.
- Packaged motors, ESCs, flight controllers, sensor modules and wiring into compact 3D-printed and CNC-machined chassis, then ran fit, clearance and acceptance testing across successive prototype builds.
- Sourced from outside vendors on small volumes and short lead times, trained the build team, and explained design decisions to customers who were not engineers.

## **ENGINEERING PROJECTS**

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### **Full Motorcycle Design & Assembly - SolidWorks**

Jul 2026 – Aug 2026

- Modeled 30 components into a single assembly with three subassemblies using solid and surface features, then produced all 30 dimensioned drawings. Top-level assembly 6.8 MB.

### **SolidWorks Flow Simulation Portfolio - Five Applied CFD Studies**

Aug 2026

- Venturi discharge coefficient validated against ISO 5167-4 within 1.16%. Counter-current heat exchanger LMTD hand-verified to 0.007%. Ventilation CO2 decay time constant within 0.83% of single-zone theory. Seven independent validations across five problems.

### **Design and Fabrication of Stair Climbing Robot - B.E. Capstone, Team of 3**

2024

- Designed the belt drive, pulleys and 6061-T6 aluminum frame in SolidWorks, sourced the bill of materials, and built the machine. Tested across four stair geometries measuring speed, power draw and stability, and traced the failure at a 45-degree rise to traction rather than motor torque, since the statics showed 2.4x margin at that point.

### **TerraExplorer - Six-Wheel Remote Sensing Rover | MJCET R&D Cell**

Oct 2023 – May 2024

- Only mechanical engineer on a four-person electronics team. Designed the chassis and rocker-bogie suspension in SolidWorks around 20x20mm aluminum T-slot profile and 3D-printed joints, then printed and assembled the running vehicle carrying an FPV camera and RF link. Received the R&D Cell's Letter of Appreciation.

### **Kinematic Gear Drive Vehicle (KGDV) | NYU Tandon**

May 2026

- Designed a fully 3D-printed multi-stage gear drivetrain at a 3:1 overall ratio, ran worst-case tolerance stack-up across the 3-stage train, specified +/-0.15mm running clearances to absorb FDM dimensional variation, and iterated 3 print cycles to a working 20-part assembly.

### **Solar Desalination Plant - CSP + Multi-Effect Distillation**

Jul 2024

- Designed a 500,000 L/day plant around a CSP power tower with 8-hour molten salt storage, ran the thermal calculations across the solar field, storage and MED train, and costed it at \$27M CAPEX against a 10-12 year break-even.

### **Commercial Retrofit HVAC Energy & Cost Analysis | NYU**

May 2026

- Applied lifecycle cost modeling to HVAC upgrade scenarios for a commercial building, comparing energy use, installation cost and Local Law 97 penalty exposure across a 15-year horizon to find the lower-NPV compliance path.

### **SAE Team Asphalt (Captain) | Club Optimus (Founder & Chairman), MJCET**

May 2021 – Jul 2024

- Finished 2nd nationally at SAE IPEC 2K23 leading a 15+ member team through design, fabrication and testing of an electric go-kart. Ran Ackermann steering geometry, braking load and structural thermal analysis in ANSYS and SolidWorks for SAE BAJA 2022. Grew Club Optimus from nothing to 100+ members across 25+ technical events.

## **CERTIFICATIONS & PUBLICATION**

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SOLIDWORKS CAD Design Associate (CSWA) - Dassault Systemes, Sep 2024

SOLIDWORKS Additive Manufacturing Associate (CSWA-AM) - Dassault Systemes, Sep 2024

SOLIDWORKS Fluid Dynamics Analysis CFD - Udemy, Omar Koryakin, Aug 2026

Mastering SOLIDWORKS (2024-25): A Complete Course - Udemy, CADCIM Technologies, Aug 2026

HVAC Troubleshooting PRO: HVAC Maintenance & Diagnosis - Udemy, ExpertEase Global, Jul 2026

Publication (first author): "Advancement in Drone Technology Integrated With Artificial Intelligence and LiDAR Sensor", IOSR Journal of Mechanical and Civil Engineering, Vol. 21(2), pp. 35-41, April 2024